KPA1500 SDR Bridge

Install and User Guide

Version 2.44

Control and watch your Elecraft KPA1500 amplifier from Thetis, Thetis HL2, SDR-VST3 or NereusSDR, from a browser panel, a small Applet window, and an optional FlexControl knob and LP-100A meter.

Written for the radio operator, not the programmer. Every step is spelled out.

KW4EX | github.com/kw4ex/kpa1500-sdr-bridge

Use at your own risk. No warranty. See Part 15.

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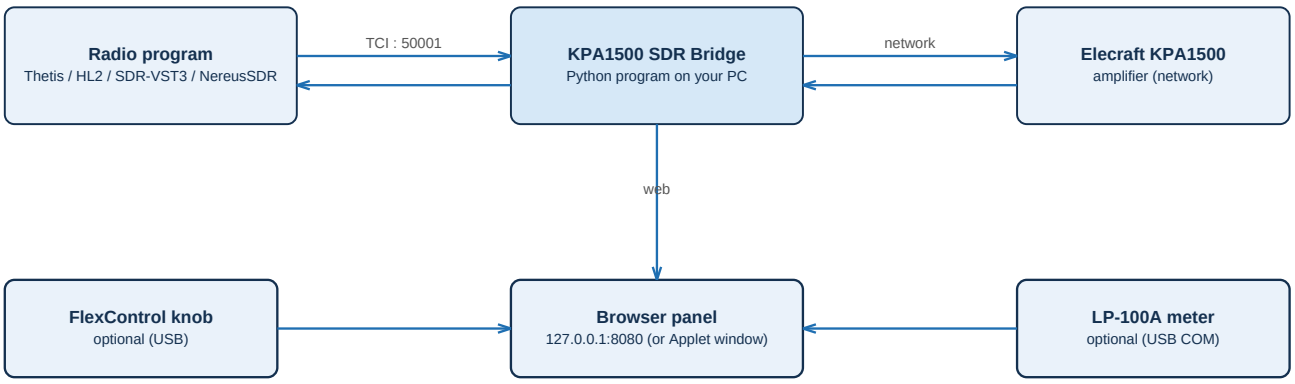
1. What the bridge does

The KPA1500 SDR Bridge is a small program that runs on your Windows PC. It sits between your SDR program and your KPA1500 amplifier. It lets the amplifier follow your radio automatically and gives you a control panel for the amplifier without needing to walk to it.

- **Follows your radio:** it reads frequency and transmit state from the SDR program, so the amplifier is always on the right band.
- **Shows the amplifier:** power, SWR, temperature, voltage, current, fan, band, ATU and fault state, live.
- **Controls the amplifier:** Operate / Standby, ATU tune, fan, power on and off, and a "safe TUNE" that puts the amplifier in Standby before the radio transmits a tune carrier.
- **Protects against no antenna:** if a band has no antenna, the bridge can warn you, put the amplifier in Standby, or refuse TUNE.
- **Three ways to see it:** a browser panel, a small Applet window, and optional windows inside Thetis (Controls, Telemetry, Knob).
- **Optional extras:** a FlexControl knob and an LP-100A watt meter.



NereusSDR also uses TCP CAT on port 13013 (frequency / TUNE).

Figure 1 - How the pieces connect.

Supported radio programs

Program	How the bridge talks to it	Notes
Thetis (standard)	TCI on port 50001	Optional Thetis windows (Controls, Telemetry, Knob).
Thetis HL2 (Hermes Lite 2)	TCI on port 50001	Detected automatically; installer sets up its windows too.
SDR-VST3	TCI on port 50001	Detected automatically; windows set up the same way.
NereusSDR	TCI plus TCP CAT (port 13013)	Needs two settings pages turned on. See Part 6.

The bridge works with one radio program at a time per amplifier, but one bridge serves all of them: you choose which program(s) you use during install, and you can switch later.

2. What you need

Item	Requirement
Computer	Windows 10 or 11 (Intel/AMD or ARM, including Windows 11 ARM under Parallels on a Mac). Any machine that can run your radio program is fine.
Python	Python 3.9 or newer. The installer installs it for you if it is missing (it uses the Windows package tool, winget).
Amplifier	Elecraft KPA1500 on your network (Ethernet). Know its IP address, or let the installer search.
Radio program	Thetis, Thetis HL2, SDR-VST3 or NereusSDR, with its TCI server (and for NereusSDR, TCP CAT) available.
Internet	Needed once for the download and for Python and pyserial. Not needed to run.
Optional	FlexControl knob; LP-100A watt meter on a USB COM port.

Before you start

1. Make sure the amplifier is powered and shows a network address (or note the address from its menu).
2. Know which radio program(s) you run.
3. Close the old version of the bridge if you have one running (the installer can stop it for you).
4. Have about 10 minutes. The self-test inside the installer takes about 2 minutes.

Tip: Open the amplifier menu and turn NET WAKE on LAN = ON if you want the bridge to be able to power the amplifier on from the network.

3. Installing with the installer (recommended)

3.1 Download

1. Go to **github.com/kw4ex/kpa1500-sdr-bridge** and click **Releases** on the right.
2. Under the newest release (v2.44), download **Install_KPA1500_SDR_Bridge.bat**. The same page has this guide's companion Help file and a zip with every file.
3. Save it somewhere easy, such as your Downloads folder.

3.2 Run it

1. Double-click **Install_KPA1500_SDR_Bridge.bat**.
2. If Windows shows a blue **Windows protected your PC** screen, click **More info**, then **Run anyway**. This is normal for any program that is not from a big publisher. The installer is a plain text file; you can open it in Notepad to see what it does.
3. A black window opens and asks questions. Press the number or letter shown, then Enter. Pressing Enter alone takes the suggested default.

3.3 The eight steps

Step	What it asks / does	What to answer
1. Python	Checks for Python and pyserial (needed for the LP-100A). Installs what is missing.	Let it install. If it says it needs to restart the window, run the installer again.
2. Install folder	Where to put the program. Default is C:\RadioScripts\KPA1500_SDR_Bridge. On an upgrade it offers **Keep your current start options**.	Press Enter for the default. On an upgrade, keeping your options is usually right.
3. Finding the amplifier	1 = search the network, 2 = type the address, 3 = skip.	Try 1. If it is not found, choose 2 and type the amplifier's IP.
4. Radio program	1 = Thetis family (Thetis, HL2, SDR-VST3), 2 = NereusSDR, 3 = Both.	Choose what you run. If you pick NereusSDR, the installer shows the NereusSDR settings you must make (also in Part 6).
5. LP-100A	Asks for its COM port, or skip.	Type the COM number (for example COM5) or skip if you have no LP-100A.
6. Browser panel / other devices	Whether other computers or phones may open the panel. Optionally sets a password.	Say No if you only use this PC. Say Yes for tablets or other computers; set a password (see Part 14).
7. Shortcuts and tests	Creates Desktop shortcuts, asks for firewall permission, runs a ~2 minute self-test, asks about starting at sign-in.	When Windows asks permission (UAC) to open network ports, click **Yes**. Let the self-test finish.
8. Thetis windows	For Thetis, HL2 and SDR-VST3: offers to add the Controls, Telemetry and Knob windows to Thetis.	Close Thetis first, then say Yes. A backup is made. See Part 7.

3.4 When it finishes

The last screen lists what was installed and repeats any NereusSDR settings. Your start shortcut is on the Desktop. Upgrading later is the same: run the newest installer; your settings are kept.

> **Safe to upgrade:** The installer never deletes your station settings. It writes a backup of the previous program files before replacing them.

4. Installing by hand (without the installer)

Use this only if you cannot run the installer.

1. Install Python 3.9 or newer from python.org. Tick **Add python.exe to PATH**.
2. Open a Command Prompt and run: `pip install pyserial`
3. Make a folder, for example `C:\RadioScripts\KPA1500_SDR_Bridge`.
4. Download the zip **KPA1500_SDR_Bridge_v2.44_files.zip** from Releases and extract all 13 files into that folder.
5. Edit **start_kpa_sdr_bridge.bat** in Notepad and put your amplifier's address after `--amp`.
6. Double-click **start_kpa_sdr_bridge.bat**. Then open a browser to `http://127.0.0.1:8080`.

The 13 files

File	Purpose
kpa1500_sdr_bridge.py	The bridge program itself.
kpa1500_thetis_setup.py	Adds / removes the windows inside Thetis.
start_kpa_sdr_bridge.bat	Starts the bridge with your options (edit this to change them).
Stop_Bridge.bat	Stops a running bridge.
Open_Applet.bat	Opens the Applet window.
Setup_Thetis_Containers.bat / Undo_Thetis_Containers.bat	Add or remove the Thetis windows.
kpa_netcheck.ps1	Network checker for troubleshooting.
KPA1500_Bridge_Help.html	Built-in Help (also opens from the panel).
README_SDR_Bridge.txt	Plain-text readme.
Install_KPA1500_SDR_Bridge.bat	The installer.

5. Starting and stopping

Start

- Double-click the **KPA1500 SDR Bridge** shortcut on the Desktop (or **start_kpa_sdr_bridge.bat**). A black window stays open while the bridge runs. That window is the bridge; do not close it unless you want to stop it.
- To start at every Windows sign-in, answer Yes to the autostart question in the installer. To undo it, delete the **KPA1500 SDR Bridge** shortcut from the Windows Startup folder (press Win+R, type `shell:startup`).

Stop

- Double-click **Stop_Bridge.bat**, or
- In the Applet window use **Bridge > Quit bridge**, or
- Close the black window.

The bridge allows only one copy at a time. Starting it again simply replaces the older copy (the start file uses `--replace`).

First-run checklist

1. Start your radio program first, and switch its TCI server on (Part 6).
2. Start the bridge. In its black window you should see it connect to the amplifier and to the radio.
3. Open `http://127.0.0.1:8080`. The amplifier tiles should show live values and the radio badge near the top should name your program.
4. Change frequency in the radio program. The bridge's band tile should follow within a second.
5. With the amplifier in Standby, try the panel's Operate button and watch the amplifier respond.

Not working? If the panel says Offline or shows no radio, jump to Part 13 (Troubleshooting).

6. Radio program settings

6.1 Thetis, Thetis HL2 and SDR-VST3

1. Open **Setup**, then **Serial/Network/Midi CAT**, then the **Network** tab.
2. In the **TCI Server** box, set the address to `127.0.0.1` and port `50001`.
3. Press the button so it shows **TCI Server Running**.
4. Leave everything else as it is.

Thetis HL2 is started with its own shortcut that includes `-dbid:HL2`. The bridge detects the HL2 and SDR-VST3 installations by their folders and databases, so no extra setting is needed.

6.2 NereusSDR

NereusSDR needs two settings pages. Both must be on.

Page	Setting	Value
TCI	Enable TCI Server	On, address 127.0.0.1, port 50001
TCI	Compatibility, VFO Quirks, IQ Stream, Sensors	Leave at defaults
TCP/IP CAT	Enable TCP CAT	On
TCP/IP CAT	CAT 1	127.0.0.1, port 13013
TCP/IP CAT	VFO A	Bound to Slice A
TCP/IP CAT	VFO B	Leave empty

To test CAT, open the **CAT Tester** in NereusSDR and send `ZZVN;`. It should answer with a version. The bridge sends ZZVN; itself every 10 seconds as a keep-alive.

Important: For TUNE and transmit control, NereusSDR needs its **TXcontrol** permission enabled. Without it the bridge can read frequency but cannot start a tune carrier.

6.3 Running the radio program on another computer

If your radio program is on a different machine, change `127.0.0.1` to that machine's address in the bridge's start options (the installer asks). On that machine, allow the TCI port through its firewall.

7. The windows inside Thetis (Thetis, HL2, SDR-VST3)

These are optional. The setup adds three small windows (containers) inside Thetis so you can see the amplifier without leaving the radio program.

Window	What it shows
Controls	Buttons: Operate/Standby, ATU, TUNE, fan, power, and an optional Title line. Button style: Standard or Legacy.
Telemetry	Power, SWR, temperature, voltage, current. Has its own option to hide the title line.
Knob	Settings for the FlexControl knob and an LP-100A meter window.

Adding and removing

1. Close Thetis completely.
2. Run **Setup_Thetis_Containers.bat** (the installer does this for you in Step 8).
3. Start Thetis. The windows appear where you had them docked or floating; drag them where you like.
4. To remove them later, close Thetis and run **Undo_Thetis_Containers.bat**. The setup always saves a backup of your Thetis database first, separately for Thetis, HL2 and SDR-VST3.

Button style (Standard or Legacy)

In the same section, next to Columns, the **Button style** box sets how the Controls window buttons look. **Standard** (the default) is black buttons that turn blue when on, the look the windows always had. **Legacy** is near-black rounded buttons with a thin grey border and a matching dark window background, like the Container Manager's legacy look. After changing it, press **Rebuild Thetis windows**. The panel's Thetis look follows the choice at once.

Choosing which buttons the Controls window shows

Open the browser panel, go to **Settings**, then the **Thetis windows** section. Tick the buttons you want in the Controls window, including **Title** (the heading line), then apply. The Telemetry window has its own title choice.

> **Note: C:\RadioScripts\ThetisContainers** is a separate container project and is never touched by this setup.

8. The browser panel

Open `http://127.0.0.1:8080` in any browser on the same PC. The panel is a set of tiles. At the top is a radio badge, showing which program the bridge is connected to.

Settings

The **Settings** button opens a drawer. The main groups are Amplifier (address and Find amplifier), the Layout bar, Tile options, and the Thetis windows (which buttons, their order, columns and Button style).

The Layout bar

Control	What it does
Layout	Pick a saved layout. **Default** is the starting one.
Save as...	Save the current arrangement under a name. **Delete** removes it.
Apply preset...	One-step looks: **Full** (everything), **Compact**, **Telemetry**, **Controls**, **Thetis look**.
Style: Tiles / Thetis look	Two overall appearances.
Size: Tile width, Labels (Short/Full), Bars	Make tiles bigger or smaller, shorten labels, show or hide bars.
Theme: Auto / Light / Dark	Follow Windows, or force light or dark.
Title bar	Show or hide the title bar.
Show all tiles	Bring back any tile you hid.
Export / Import	Save your layouts to a file or load them on another computer.

Layout web addresses

Add `?layout=Name` to the address to open a saved layout directly, for example `http://127.0.0.1:8080/?layout=Compact`. This is handy for bookmarks: one bookmark per look, or a different layout on a tablet.

Rearranging

Tiles can be dragged into a new order, hidden, and resized from the Settings drawer. Your changes are saved in the layout you are using.

9. The Applet window

The Applet is a small always-handy window with the same information. It opens from the panel (or by double-clicking **Open_Applet.bat**), and is at `http://127.0.0.1:8080/applet`.

The Look menu

Item	Use
Theme / Colours	Light, dark or custom colours.
Size and spacing	Make it larger or tighter.
Top line: show	Show or hide the top status line.
Sections: show and order	Choose which sections appear and in what order.
Meters / Buttons	Pick which meters and buttons are present.
Window title	Set the window's title text.
Resize window	Set an exact size.
Safety: over-drive protection	Limit in watts and Ignore / Warn / Hold at the limit. See Part 11b.
Safety: no-antenna check	Only warn; Warn and put amp in STANDBY; or Refuse TUNE and warn. See Part 11.
Bridge > Quit bridge	Stops the bridge.
Share or restore a look	Copy your look to another computer or bring it back.
Radio program	Which radio program the Applet reports against.

10. Operating the amplifier

Operate and Standby

The big Operate / Standby button changes the amplifier state. The amplifier must be in Operate to amplify; in Standby it passes the signal straight through.

Safe TUNE

Tuning with the amplifier in Operate can stress it. The safe TUNE button first puts the amplifier in **STANDBY**, then starts the radio's tune carrier, and after TUNE ends returns the amplifier to its previous state. In Thetis family programs this uses the radio's TUNE; in NereusSDR it uses CAT (ZZTU1; to start, ZZTU0; to stop).

ATU, fan and power

- **ATU**: starts the amplifier's internal tuner.
- **Fan**: sets fan speed.
- **Power**: powers the amplifier off, or on if Wake-on-LAN is enabled in the amplifier menu.

Standby on Tune

This option makes the bridge put the amplifier in Standby whenever you press the radio's own TUNE button, not just the bridge's. It is set in the Applet and panel settings.

Remember: The bridge is a convenience. It is not a substitute for the amplifier's own protection, and you remain responsible for safe operation.

11. No-antenna protection

If a band has no antenna connected, transmitting into the amplifier can be harmful. The bridge remembers which bands have no antenna and acts when you go to one.

Choose what it does (Applet: **Look > Safety: no-antenna check**):

- **Only warn** - shows a warning.
- **Warn and put amp in STANDBY** - also forces Standby.
- **Refuse TUNE and warn** - blocks the safe-TUNE button on that band.

Teaching the bridge a no-antenna band (Learn)

1. Go to the band in your radio program.
2. **Disconnect the antenna** from the amplifier output, or select the position that has no antenna.
3. Wait about 3 seconds so the amplifier settles.
4. Press **Learn this band**. The bridge records the reading for that band.
5. Reconnect your antenna. Repeat for any other band that has no antenna.

Careful: Learn must be done with the antenna **disconnected**. Learning with the antenna connected will teach the wrong reading.

11b. Over-drive protection

Too much drive can damage the amplifier. This option lets you set **one limit, in watts, for the drive going into the KPA1500** and choose what the bridge does when the drive goes over it. It works the same with Thetis, Thetis HL2, SDR-VST3 and NereusSDR.

- **Before you transmit:** the bridge reads the drive setting (the drive slider) of your radio program about once a second.
- **While you transmit:** the bridge also watches the drive it actually measures going into the amplifier.
- **Only in OPERATE:** with the amplifier in STANDBY the signal passes straight through, so the check waits. This lets you tune the amplifier with its own ATU, left in STANDBY, at full radio power. The check also waits while the amplifier is tuning with its ATU, and TUNE with Standby-on-Tune is not limited by it.
- **Short peaks are ignored:** the measured drive must stay over the limit for about 0.8 seconds before it counts. The warning then stays on screen for 3 seconds.

Settings

Setting	Meaning
Limit (W)	The most drive, in watts, you want going into the amplifier. Empty or off switches the protection off.
If it goes over	**Ignore** does nothing. **Warn** shows a red message. **Hold at the limit** shows the message and puts the radio's drive setting back to the limit before you transmit. If the measured drive still stays over the limit for more than a second while transmitting, the amplifier is put in STANDBY until you lower the drive and press OPERATE.
Radio full power (W)	The slider is a percentage of the radio's full power. 100 for a 100 W radio (the default), 5 for a Hermes Lite 2. Slider 80 on a 100 W radio is 80 W.

Where to set it

1. Panel: open **Settings**, find **Over-drive protection**, type the limit and choose what to do.
2. Applet: open the **Look** menu, section **Safety: over-drive protection**.

Remember: The measured drive has a 10 percent (plus 1 W) margin so the amplifier's own reading does not cause false alarms. This helps; it does not replace care. The amplifier's own protection stays the last line of defence.

12. FlexControl knob and LP-100A

FlexControl knob

If you have a FlexControl, the bridge can use its knob and three buttons to control the radio program. Turn it on in the Knob window or the panel settings. The knob turns have **modes**; the presses run actions.

Knob turn modes

VFO A tuning, VFO B tuning, AF volume (master), AGC gain (RX1), RIT (10 Hz per click), XIT (10 Hz per click), Tuning step size, TX drive.

Default press actions

Press	Action
Knob: tap	AF volume / VFO A (toggle)
Knob: double tap	Mute on / off
Knob: long	Back to VFO A tuning
Button 1: tap	AF volume / VFO A (toggle)
Button 1: double tap	AGC gain / VFO A (toggle)
Button 1: long	Mute on / off
Button 2: tap	VOX on / off
Button 2: double tap	Tuning step: larger
Button 2: long	Tuning step: smaller
Button 3: tap	Safe TUNE (amp STANDBY first) / stop
Button 3: double tap	Band up
Button 3: long	Band down

Each press can be reassigned from the Knob settings.

LP-100A watt meter

Plug the LP-100A in by USB and note its COM port (Windows Device Manager, Ports). Give that port to the installer (Step 5) or put it in the start options. The bridge reads it with pyserial. Set the LP-100A to its **Watts** screen. The default speed is 115200 baud; units with firmware older than 1.2.0.0 use 38400.

13. Troubleshooting

Problem	Try this
Panel will not open	Is the black bridge window running? If not, start it. Check the address is exactly http://127.0.0.1:8080.
Panel shows amplifier offline	Check the amplifier's IP in Settings and use **Find amplifier**. Make sure the PC and the amplifier are on the same network.
No radio / badge is empty	Start the radio program and turn its TCI server on (Part 6). Confirm port 50001.
Frequency does not follow	For NereusSDR, check the TCP CAT page: VFO A bound to Slice A. Test with ZZVN; in the CAT Tester.
TUNE does nothing in NereusSDR	Enable TXcontrol in NereusSDR.
Another computer cannot open the panel	Bridge must be started with `--web-bind 0.0.0.0` and Windows firewall must allow port 8080 (the installer does this). See Part 14.
Bridge says it is already running	Use Stop_Bridge.bat, or just start again (the start file replaces the old copy).
Thetis windows missing after an update	Close Thetis and run Setup_Thetis_Containers.bat again.
LP-100A shows nothing	Check the COM number, close other programs that use it, set the meter to the Watts screen, check baud rate.
Installer window closes at once	Right-click the file, Run as administrator, or run it from a Command Prompt to read any message.
A Thetis button does nothing, but the meters work	Windows may have reserved the port Thetis uses for buttons. The bridge shows a yellow PORT message when it starts. Fix: close Thetis, run Setup_Thetis_Containers.bat (it moves the buttons to a free port), start the bridge, then Thetis. To check by hand, in Command Prompt: `netsh int ipv4 show excludedportrange protocol=udp`. The Thetis button port (47102, or 50102 on older installs) must not be inside a listed range.
Over-drive protection says it is waiting	It only works while the amplifier is in OPERATE. In STANDBY the limit is not applied.
Want to check the network	Run kpa_netcheck.ps1 (right-click, Run with PowerShell). It tests the amplifier, radio and ports.

14. Using the panel from another device

You can open the panel from a tablet, phone or another computer.

1. Start the bridge with `--web-bind 0.0.0.0` (the installer asks in Step 6).
2. Allow port 8080 in Windows firewall (the installer does this with a UAC prompt).
3. Find the bridge PC's address (Command Prompt, `ipconfig`), for example 192.168.1.50.
4. On the other device open `http://192.168.1.50:8080`.

Password

Set a panel password with `--web-token YourPassword` (or in the installer). Then open `http://192.168.1.50:8080/?t=YourPassword`. Use a password you do not use anywhere else; it travels unencrypted on your home network.

Windows under Parallels on a Mac

Set the virtual machine's network to **Bridged** so other devices can reach it. With Shared networking only the Mac can see it.

From outside your home

Do not open port 8080 to the Internet. The best free option is **Tailscale** (free Personal plan, up to 6 users, unlimited devices, non-commercial). Install it on the bridge PC and on your laptop or phone, sign in to the same account, then use the PC's Tailscale address (100.x.x.x) in place of the home address. Nothing is opened on your router.

15. Updating, going back, uninstalling, safety

Updating

Download the newest installer from Releases and run it. Choose to keep your current start options. A backup of the previous program is saved.

Going back

Backup copies of replaced files end in `.bak` in the install folder. Rename one over the new file, or re-run an older installer from the Releases page.

Uninstalling

Double-click **Uninstall_KPA1500_SDR_Bridge.bat** in the install folder. It stops the bridge, offers to take the KPA1500 windows out of Thetis (close Thetis first), removes the shortcuts, the firewall rules and the saved Applet window position, and, if you say yes, the program files. Nothing is deleted without asking.

Mac and Linux (experimental)

A separate installer, `Install_KPA1500_SDR_Bridge.sh`, runs the bridge on a Mac or Linux computer (for example with NereusSDR). Copy it and README_Mac_Linux.txt to the computer, open Terminal in that folder and type `bash Install_KPA1500_SDR_Bridge.sh`. Everything goes in your own folders, no administrator password is needed, and `uninstall_kpa1500_sdr_bridge.sh` removes it completely. It has been tested on Linux only; the Thetis container windows are Windows-only.

Safety and no warranty

This is amateur radio software provided free, as is, with no warranty of any kind. You are responsible for your station and your amplifier. Always watch the amplifier, keep its own protection active, and never rely on any software as your only safeguard. Elecraft, Thetis, HL2, SDR-VST3, NereusSDR and LP-100A are names of their respective owners; this project is independent of them.

Appendix A - Ports

Port	Use
8080	Bridge browser panel and Applet
50001	TCI (all four radio programs)
13013	NereusSDR TCP CAT
50110 (UDP)	NereusSDR containers (JSON)
50100 / 50101 (UDP)	Thetis MMIO (bridge listens on 50101)
47102 (UDP)	Thetis button sender (new installs; older ones 50102). Keep it below 49152, where Windows does not reserve ports

Appendix B - Where things are

Item	Location
Program and settings	C:\RadioScripts\KPA1500_SDR_Bridge

Item	Location
Start file (edit options)	start_kpa_sdr_bridge.bat in that folder
Built-in Help	KPA1500_Bridge_Help.html in that folder
Project page and downloads	github.com/kw4ex/kpa1500-sdr-bridge